COMPSCI 713: AI Fundamentals

Lecture 5: Knowledge Representation

Fundamentals and Methods

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What You Will Learn Today?

- Introduction to Knowledge Representation
- Symbolic Logic in Knowledge Representation
- Semantic Networks in Knowledge Representation
- Frames in Knowledge Representation
- Rule-Based System in Knowledge Representation
- Knowledge Graphs in Knowledge Representation

What is Knowledge Representation?

Definition:

Knowledge Representation (KR) refers to the methods used in AI to **store**, **retrieve**, and **handle** knowledge to enable intelligent reasoning.

Why Do We Need KR?

- Bridges raw data & intelligent decision-making
- Allows AI to reason logically and infer new facts
- Enables knowledge-driven AI applications (e.g. expert systems, search engines, autonomous robots, chatbots)

Example: AI in Medical Diagnosis

- A medical AI system stores knowledge about symptoms, diseases, and treatments.
- Given a patient's symptoms, the system can infer the most likely disease and suggest treatments.

Why Does AI Need Knowledge Representation?

AI Needs Structured Knowledge to Perform Intelligent Tasks

- AI systems rely on knowledge to understand, reason, and make decisions.
- Without structured knowledge, AI models would only work with raw data and struggle with logical reasoning.

Real-World AI Applications Using Knowledge Representation

Application	How It Uses Knowledge Representation
Google Search	Uses a **Knowledge Graph** to organize and retrieve relevant information.
IBM Watson	Uses **structured knowledge** for question-answering and medical diagnosis.
Autonomous Vehicles	Represents **traffic rules, objects, and road conditions** for safe navigation.
AI Chatbots (e.g. ChatGPT, Siri)	Stores **linguistic and factual knowledge graphs** for better responses.
Recommendation Systems (e.g. Netflix, Amazon)	Uses **semantic networks and knowledge graphs** to suggest relevant content.

Quick Questions: Any Other?

Structured Knowledge vs. Unstructured Knowledge

** **What is Structured Knowledge?**

Definition:

- Knowledge that is organized in a predefined format with a clear structure.
- Can be stored in databases, tables, ontologies, and knowledge graphs.

Examples:

- **Relational Databases**: Tables with rows and columns (e.g., SQL databases).
- **Knowledge Graphs**: Representing entities and relationships (e.g., Google Knowledge Graph).
- **Ontologies**: Hierarchical classification of concepts (e.g., medical taxonomy).

** **What is Unstructured Knowledge?**

Definition:

- Knowledge that does not follow a predefined structure.
- Found in raw text, images, videos, and free-form documents.

Examples:

- **Text Data**: News articles, research papers, social media posts.
- **Multimedia Content**: Videos, images, audio recordings.
- **Human Communication**: Conversations, emails, transcripts.

Structured Knowledge vs. Unstructured Knowledge

Comparison Table

Feature	Structured Knowledge	Unstructured Knowledge
Format	Organized in tables, graphs, or schemas	Free-form (text, images, videos)
Storage	Databases, ontologies, knowledge graphs	Documents, multimedia files
Processing	Fast and efficient queries	Requires NLP, deep learning
Interpretability	High – easy to understand	Low – requires advanced AI
Flexibility	Rigid – schema-dependent	Flexible – can capture complex knowledge

Exercise 1: Which of the following is an example of **structured** knowledge?

A) A collection of handwritten medical prescriptions.

B) A database storing customer purchase histories.

C) A video recording of a classroom lecture.

D) A set of research papers in PDF format.

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A) A collection of handwritten medical prescriptions.
B) A database storing customer purchase histories.
C) A video recording of a classroom lecture.
D) A set of research papers in PDF format.

Answer: B. Structured knowledge is organized in a predefined format like databases, tables, or knowledge graphs.

Five Key Requirements of Knowledge Representation (KR)

Why Do We Need Requirements for KR?

- Not all KR methods are equally effective in AI applications.
- A good KR system must balance efficiency, flexibility, and interpretability.

Requirement	Description	Example
Expressiveness	Can the KR method represent **complex** and **abstract** knowledge?	A **self-driving car** must represent traffic rules, pedestrian movement, road conditions
Computational Efficiency	Can the KR method **process information quickly**?	AI fraud detection must analyze thousands of transactions per second.
Scalability	Can the KR system **handle large and growing knowledge bases**?	Google's Knowledge Graph contains **billions** of facts and relationships.
Interpretability	Can **humans understand** how AI makes decisions?	AI medical diagnosis systems must provide **clear reasoning** for treatment recommendations.
Modifiability	Can the KR system **update itself** with new knowledge?	AI chatbots must constantly **learn from new conversations**.

Five Key Requirements of Knowledge Representation

Case Study: How a Self-Driving Car Meets These KR Requirements

🚗 How a Self-Driving Car Uses Knowledge Representation

- **Expressiveness** → Represents real-world road conditions, traffic signals, and vehicle movement.
- **Computational Efficiency** → Processes sensor data in real-time for immediate decisions.
- **Scalability** → Expands knowledge of new routes and driving patterns.
- **Interpretability** → AI must explain why it brakes or changes lanes.
- **Modifiability** → Updates driving models based on new road conditions.

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Symbolic Logic in Knowledge Representation

What is Symbolic Logic?

- **Definition**: Represents knowledge using **formal symbols** and **logical expressions**.
- Used in AI for reasoning, inference, and formal verification.
- **Types of Logic Used in KR:**
  - **Propositional Logic** – Uses simple true/false statements.
  - **First-Order Logic (FOL)** – Allows relationships between entities (e.g. "All humans are mortal").

Example: Medical Diagnosis System (FOL Rules)

- $\forall x\ (\text{Flu}(x) \rightarrow \text{HasSymptom}(x, \text{Fever}))$
  "If a patient has the flu, they will have a fever."
- $\forall x\ (\text{HasSymptom}(x, \text{Fever}) \wedge \text{HasSymptom}(x, \text{Cough}) \rightarrow \text{LikelyDiagnosis}(x, \text{Flu}))$
- AI can infer from known symptoms to diagnose patients.

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Semantic Networks in Knowledge Representation

What is a Semantic Network?

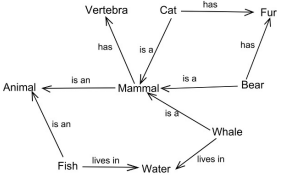
A graph-based knowledge representation where:

- **Nodes** represent concepts or objects.
- **Edges** represent relationships between them.

Used to structure knowledge hierarchically and enable inference based on relationships.

Basic Structure:

- **Nodes (Entities/Concepts):** Represents objects, ideas, or concepts (e.g. "Cat", "Mammal", "Animal").
- **Edges (Relationships/Connections):** Define how entities are related (e.g. "is-a", "has-part", "related-to").



Example Semantic Network:

- "Dog → is-a → Mammal" "Mammal → is-an → Animal" "Cat → has → Fur"

Additional Material:

`https://www.geeksforgeeks.org/semantic-networks-in-artificial-intelligence/`

`https://www.youtube.com/watch?v=dU9WsMglBSM`

Semantic Networks in Knowledge Representation

How Does Inference Work in Semantic Networks?

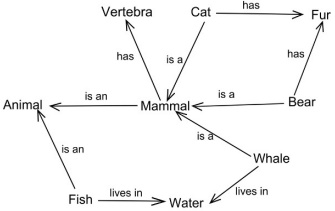
Semantic networks allow AI to **derive new knowledge** by traversing relationships.

Example: Hierarchical Reasoning (IS-A Inference)

- Given: "Cat → is-a → Mammal", "Mammal → is-an → Animal"
- AI can infer: "Dog is an Animal" (through transitive inheritance).

Example: Property Inheritance

- Given: "Mammal → has-property → Warm-blooded", "Dog → is-a → Mammal"
- AI can infer: "Dog is warm-blooded" (inherits properties from Mammal).



Semantic Networks in Knowledge Representation

Strengths of Semantic Networks

1. **Natural Representation** – Mimics human thought; relationships are intuitive and visually clear.
2. **Supports Logical Inference** – AI can deduce new facts through IS-A and HAS-PROPERTY relationships.
3. **Efficient Knowledge Retrieval** – Graph structures allow fast lookups using connected nodes.

Weaknesses of Semantic Networks

1. **Can Become Too Complex** – Large networks with millions of nodes can be hard to manage.
2. **No Standardized Representation** – Different AI models use different graph structures; integration is difficult.
3. **Poor Handling of Uncertainty** – Assumes relationships are deterministic (e.g. "Birds can fly" – but what about penguins?).

Exercise 2:

Which of the following **best demonstrates inference** in a Semantic Network?

A) A chatbot randomly generating responses.

B) AI deducing that "whales are mammals" based on an "is-a" relationship.

C) A search engine ranking web pages based on popularity.

D) A neural network recognizing images of cats.

Semantic Networks in Knowledge Representation

Exercise 2:

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D) A neural network recognizing images of cats.

Answer: B. Semantic Networks allow AI to infer properties and relationships based on hierarchical connections (e.g. IS-A and HAS-PROPERTY).

*A. A chatbot that **randomly generates responses** is not performing logical inference while in a Semantic Network, **inference** relies on **relationships between concepts**.*

*C. Search engines (like Google) rank web pages **based on user behavior** (e.g., clicks, backlinks, and SEO techniques), not by **inference** from a structured knowledge representation.*

*D. Neural networks process **unstructured data** (images, text, speech) through pattern recognition but do not inherently infer new relationships based on **existing symbolic knowledge**.*

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Frames in Knowledge Representation

What is a Frame in Knowledge Representation?

- A **frame** is a structured representation that groups related information about an entity into a **slot-filler** structure.
- A **slot** is an attribute or property that stores specific knowledge (text, lists, rules, or procedural actions).
- Inspired by human cognition: when we think about an object, we recall related attributes, actions, and relationships.

Key Concept: A frame = concept/object; a **slot** = attribute/property; a **filler** = value for that slot.

Frame	**Slot (Attribute)**	**Filler (Value)**
Car	Brand	Tesla
	Colour	Red
	Engine	Electric
	Owner	Alice

Additional Material:
https://www.geeksforgeeks.org/frames-in-ai-knowledge-representation-and-inheritance/

Frame-Based Reasoning

How Do Frames Enable AI to Reason?

Frames allow AI to infer missing knowledge using **default values**, **inheritance**, and **relationships**.

1. Default Values

- If a slot is empty, AI can use default knowledge.
- Example: Frame **Dog**, Slot **Has legs** $\rightarrow$ Default value = 4. AI infers a newly introduced dog has 4 legs unless specified otherwise.

2. Inheritance (Frame Hierarchies)

- Frames inherit attributes from higher-level frames (similar to Object-Oriented Programming).
- Example: Frame **Mammal** $\rightarrow$ Has Hair = True; Frame **Dog** (inherits from Mammal) $\rightarrow$ Has Hair = True.

Frame-Based Reasoning

3. Slot Constraints & Conditions

- Some slots have **restrictions on valid values**.
- Example: Frame **Student**, Slot **Age** $\rightarrow$ Constraint: Must be > 5 years old.

4. Procedural Attachment (Rules Triggered by Frames)

- Some slots **trigger actions when accessed**.
- Example: Frame **Hotel Reservation**, Slot **Check-in Time** $\rightarrow$ Action: Send confirmation email.

Example: AI-Assisted Personal Assistant Using Frames

Frame: Lecture	**Slot (Attribute)**	**Filler (Value)**
2026 S1 CS713 Lecture on Monday	Date	Every Monday
	Time	12:00 PM
	Location	BLT100/106-100
	Attendees	Students from CS713

Frames in Knowledge Representation

Strengths of Frames

1. **Structured & Organized** – Frames **group related knowledge** into **slot-filler structures**, making retrieval efficient. (e.g. "Restaurant" frame: Name, Location, Menu, Opening Hours).
2. **Inheritance and Default Reasoning** – AI infers missing values using **default values and hierarchical inheritance**. (e.g. *If a "Dog" frame does not specify the number of legs, it inherits "4" from the "Mammal" frame.*).
3. **Procedural Knowledge (Rules & Actions)** – Slots can trigger **actions** (procedural attachments) (e.g. A "Hotel Booking" frame triggers an "email confirmation" when a reservation is made).
4. **Easy to Update & Modify** – Slots and fillers can be modified **dynamically** (e.g. An AI assistant can update a "Lecture" frame when the time changes).

Frames in Knowledge Representation

Weaknesses of Frames

1. **Rigid Structure & Limited Flexibility** – Frames work well when knowledge is predefined; struggle with ambiguous or novel cases (e.g. A rigid "Vehicle" frame with predefined slots might not account for futuristic self-driving cars).
2. **Poor Handling of Uncertainty** – Assume knowledge is complete; difficult to reason with probabilities (e.g. A "Weather" frame may struggle to represent the uncertainty of a 70% chance of rain.).
3. **Hard to Scale for Large Knowledge Bases** – Grow complex as entities and slots increase (e.g. A healthcare system with thousands of frames (patients, symptoms, treatments) may be difficult to maintain.).
4. **Limited Logical Reasoning** – Unlike Symbolic Logic, frames do not perform deep logical deductions (e.g. Frames can store "All birds can fly," but they don't automatically reason exceptions like penguins.).

Frame-Based Reasoning – Exercise 3

Scenario: An AI-powered healthcare assistant uses **Frames** to manage patient medical records and suggest possible diagnoses based on symptoms and medical history.
Patient **Alice** reports: **fever** and **headache**.
The AI system checks **Alice's medical history** using the following frames:
Patient Frame: Alice

Frame: Patient	**Slot**	**Filler**
Alice	Age	25
	Symptoms	Fever, Headache
	Family History	None
	Recent Travel	Tropical regions (2 weeks ago)
	Vaccination History	No recent travel vaccines

Frame: Malaria	**Slot**	**Filler**
Malaria	Common Symptoms	Fever, Headache, Chills
	Transmission Risk	High in tropical regions
	Prevention	Vaccination

Frame-Based Reasoning – Exercise 3

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Question: What is the most likely reason the AI flags **Malaria** as a potential diagnosis?

A) Alice has a family history of malaria.

B) Malaria is common in all patients with pain in the lungs.

C) Malaria is common in Age from 20 to 30.

D) Alice recently traveled to a high-risk area and shows matching symptoms.

Frame-Based Reasoning – Exercise 3

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D) Alice recently traveled to a high-risk area and shows matching symptoms.

Answer: D. Alice has two key symptoms matching Malaria (fever & headache); she traveled to a malaria-prone region two weeks ago (within incubation period); she has not received a malaria vaccine, increasing susceptibility.

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Rule-Based System in Knowledge Representation

What is a Rule-Based System in AI?

- A Rule-Based System (RBS) represents knowledge as a set of **IF-THEN** rules that trigger actions when conditions are met.
- Focus on explicit **decision-making** (unlike Frames or Semantic Networks).

Structure: IF (Condition) → THEN (Action/Conclusion). AI checks facts and applies the appropriate rule.
Used in: expert systems, diagnostics, fraud detection, automated decision-making.

Rule ID	IF Condition	THEN Conclusion
R1	Fever AND Cough	Possible Diagnosis = Flu
R2	Fever AND Joint Pain AND Travel to Tropical Area	Possible Diagnosis = Dengue Fever
R3	Cough AND Difficulty Breathing	Possible Diagnosis = Pneumonia

Inference: *Fever + Cough + Joint Pain + Recent Travel ⇒ "Flu OR Dengue Fever" (R1 and R2).*

Rule-Based System in Knowledge Representation

Strengths of Rule-Based Systems

1. **Transparent and Explainable** – Every decision is based on clear, human-readable IF-THEN rules (e.g. An AI doctor follows predefined rules for diagnosis, making its reasoning understandable to doctors.).
2. **Easy to Implement for Well-Defined Problems** – Works effectively in structured domains with known rules (e.g. fraud detection system: "IF transaction $> \$10{,}000$ AND foreign country, THEN flag as suspicious").
3. **Works Without Large Training Data** – Unlike ML, does not require massive datasets (e.g. A legal AI assistant can apply IF-THEN logic to analyze contract clauses without needing prior training.).

Rule-Based System in Knowledge Representation

Weaknesses of Rule-Based Systems

1. **Hard to Scale with Complex Knowledge** – As rules increase, managing thousands of IF-THEN rules becomes difficult. (e.g. A tax AI needs thousands of rules for different tax laws, making updates challenging).
2. **Poor Adaptability to New Situations** – Cannot generalize beyond predefined rules (e.g. An AI chatbot using rules may fail when users ask unexpected questions).
3. **Requires Expert Knowledge to Define Rules** – Rules must be handcrafted by domain experts (e.g. A legal AI must be programmed with thousands of rules by lawyers).

Additional Material: `https://www.geeksforgeeks.org/rule-based-system-in-ai/`
`https://www.youtube.com/watch?v=BXHcPESoaPY`

Rule-Based System – Exercise 4

Scenario: Rule-Based AI system for **fire detection** in a smart building. IF-THEN rules trigger alarms based on sensor readings.
The AI has the following rules:

Rule ID	IF Condition	THEN Conclusion
R1	Temperature $> 60°C$ **AND** Smoke Detected	Trigger Fire Alarm
R2	Temperature $> 80°C$	Trigger Emergency Evacuation
R3	Carbon Monoxide $>$ Safe Limit	Alert Building Manager
R4	Sprinklers Activated **AND** Smoke Detected	Confirm Fire

Current Sensor Readings:

- Temperature $= 85°C$ Smoke Detected = YES Carbon Monoxide = Safe Sprinklers Activated = NO

Question: What actions will the AI take?

- A) Trigger Fire Alarm and Emergency Evacuation.
- B) Only Alert the Building Manager.
- C) Only Trigger the Fire Alarm.
- D) Confirm Active Fire and Trigger Evacuation.

Rule-Based System – Exercise 4

Answer: A – Trigger Fire Alarm and Emergency Evacuation.

Reasoning:

- **R1** fires: Temperature $> 60^\circ C$ AND Smoke Detected $\Rightarrow$ **Fire Alarm** triggered.
- **R2** fires: Temperature $> 80^\circ C$ $\Rightarrow$ **Emergency Evacuation** triggered.
- **R3** does NOT fire: Carbon Monoxide is at safe level.
- **R4** does NOT fire: Sprinklers are NOT activated, so AI cannot confirm active fire yet.

Final AI Actions: Trigger Fire Alarm (R1); Trigger Emergency Evacuation (R2).

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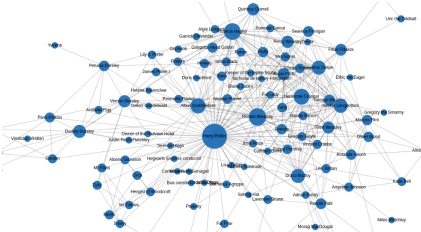
Knowledge Graphs in Knowledge Representation

What is a Knowledge Graph?

- A **graph-based** representation that connects **entities** (nodes) with **relationships** (edges).
- Unlike Frames or Rule-Based Systems, a Knowledge Graph
  - Stores structured knowledge (facts, concepts, relationships)
  - Enables AI to infer new connections from known relationships
  - Provides a scalable and flexible way to organize information.

Components:

- **Nodes** = Entities (people, places, objects)
- **Edges** = Relationships
- **Properties** = Attributes of nodes



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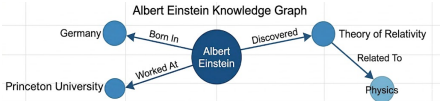

Entity (Node)	Relation (Edge)	Entity (Node)
Albert Einstein	Born In	Germany
Albert Einstein	Discovered	Theory of Relativity
Theory of Relativity	Related To	Physics
Albert Einstein	Worked At	Princeton University

Inference: If we know "Theory of Relativity is related to Physics" and "Einstein discovered the Theory of Relativity", then AI can infer: "Einstein contributed to Physics."

Knowledge Graphs in Knowledge Representation

How Does AI Use Inference in Knowledge Graphs?

Knowledge Graphs allow AI to infer new facts by analyzing existing relationships.

1. **Transitive Inference** ($A \to B$, $B \to C \Rightarrow A \to C$)
   Example: "Albert Einstein discovered Theory of Relativity"; "Theory of Relativity is part of Physics" $\Rightarrow$ "Einstein contributed to Physics."
2. **Relationship Expansion** – Identify hidden connections
   Example: If two lectures worked on the same course, AI can infer a collaboration link.
3. **Entity Disambiguation** – Helps AI distinguish entities with similar names.
   Example: "Apple (Company)" vs. "Apple (Fruit)".
4. **Question Answering** – Retrieve structured answers.
   Example: "Who invented the telephone?" $\Rightarrow$ Alexander Graham Bell. (from graph relations)

Knowledge Graphs in Knowledge Representation

Strengths of Knowledge Graphs (KG)

1. **Highly Structured & Interpretable** – Unlike unstructured text, a Knowledge Graph provides clearly defined relationships between entities.
   Example: *Google's KG organizes people, places, and facts in a structured way.*
2. **Enables Inference & Knowledge Discovery** – Infer missing knowledge based on known relationships
   Example: *If a scientist worked on a theory, AI can predict possible collaborations.*
3. **Scalable for Large-Scale Knowledge Representation** – Works well with millions of facts and relationships.
   Example: *AI assistants (Siri, Alexa) retrieve structured knowledge in real time.*
4. **Supports Multi-Domain Knowledge Integration** – Combine medical, scientific, business, and general knowledge into one system. Example: *Medical AI links symptoms, diseases, treatments, and drugs for diagnos.*

Knowledge Graphs in Knowledge Representation

Weaknesses of Knowledge Graphs

1. **Incomplete Knowledge & Data Sparsity** – If information is missing, AI cannot infer accurate answers.
   Example: *If a Knowledge Graph lacks data about a rare disease, the AI cannot make correct predictions.*
2. **High Maintenance & Complexity** – Requires constant updates to add new entities and relationships.
   Example: *Keeping a Knowledge Graph updated for AI chatbots like ChatGPT requires continuous fact-checking.*

Additional Material: `https://docs.stardog.com/getting-started-series/getting-started-1`
`https://www.youtube.com/watch?v=y7sXDpffzQQ`

Knowledge Graphs – Exercise 5

Scenario: An AI system uses a Knowledge Graph to organize historical figures, scientific discoveries, and their impact. The system uses relationship-based inference to answer queries.

The Knowledge Graph contains the following known relationships:

Entity (Node)	Relation (Edge)	Entity (Node)
Isaac Newton	Discovered	Law of Gravity
Law of Gravity	Related To	Physics
Albert Einstein	Contributed To	Physics
Albert Einstein	Developed	Theory of Relativity
Theory of Relativity	Related To	Gravity
Theory of Relativity	Influenced By	Law of Gravity

Question: Did Isaac Newton's discoveries influence Albert Einstein?

- A) No, no direct link between Newton and Einstein.
- B) Yes, Einstein contributed to Physics, and Physics includes Gravity.
- C) Yes, Newton discovered Law of Gravity, and Theory of Relativity was influenced by it.
- D) No, Einstein worked on different theories.

Knowledge Graphs – Exercise 5

Answer: C – Yes, because Newton discovered the Law of Gravity, and the Theory of Relativity was influenced by it.

Reasoning:

- The graph does not *directly* connect Newton to Einstein, but AI can infer by **traversing** the graph:
- Newton → Discovered → Law of Gravity
- Law of Gravity → Influenced → Theory of Relativity
- Einstein → Developed → Theory of Relativity
- So Einstein's theory was influenced by Newton's discovery.

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Theory of Relativity	Related To	Gravity
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Relationship Between KR Methods

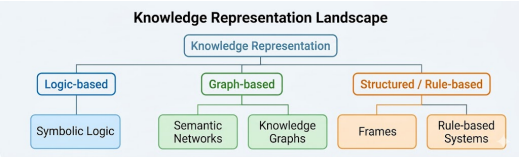
How are these Knowledge Representation methods related?

- Symbolic Logic, Semantic Networks, Frames, Rule-Based Systems, and Knowledge Graphs are all **different approaches** to Knowledge Representation.
- They are **not** subtypes of one another; instead, they represent knowledge in **different ways** and support **different kinds of reasoning**.

A simple way to organize them:

Key idea:

- **Semantic Networks** and **Knowledge Graphs** both use graphs.
- **Frames** organize knowledge using slot–filler structures.
- **Rule-Based Systems** use IF–THEN rules for decisions.
- **Symbolic Logic** uses formal statements for precise inference.



Conclusion: They are **different KR paradigms**, not different types of Knowledge Graphs.

Summary – Knowledge Representation (KR)

What have we learned today?

- KR enables AI systems to **store, organize, retrieve, and reason** over knowledge.
- Different KR methods provide different ways to represent knowledge, depending on the task and application.

Main KR methods covered today:

- **Symbolic Logic** – represents knowledge using formal logical expressions for precise reasoning.
- **Semantic Networks** – represent concepts and relationships as nodes and edges in a graph.
- **Frames** – organize knowledge into slot–filler structures for describing entities and their attributes.
- **Rule-Based Systems** – use IF–THEN rules for explicit decision-making and inference.
- **Knowledge Graphs** – connect entities, relationships, and properties in a scalable graph structure.

Key takeaway:

- There is **no single best KR method**.
- Different methods have different strengths in **representation, inference, scalability, and interpretability**.
- In practice, AI systems may combine multiple KR methods to solve real-world problems.

Thank you!

Q&A